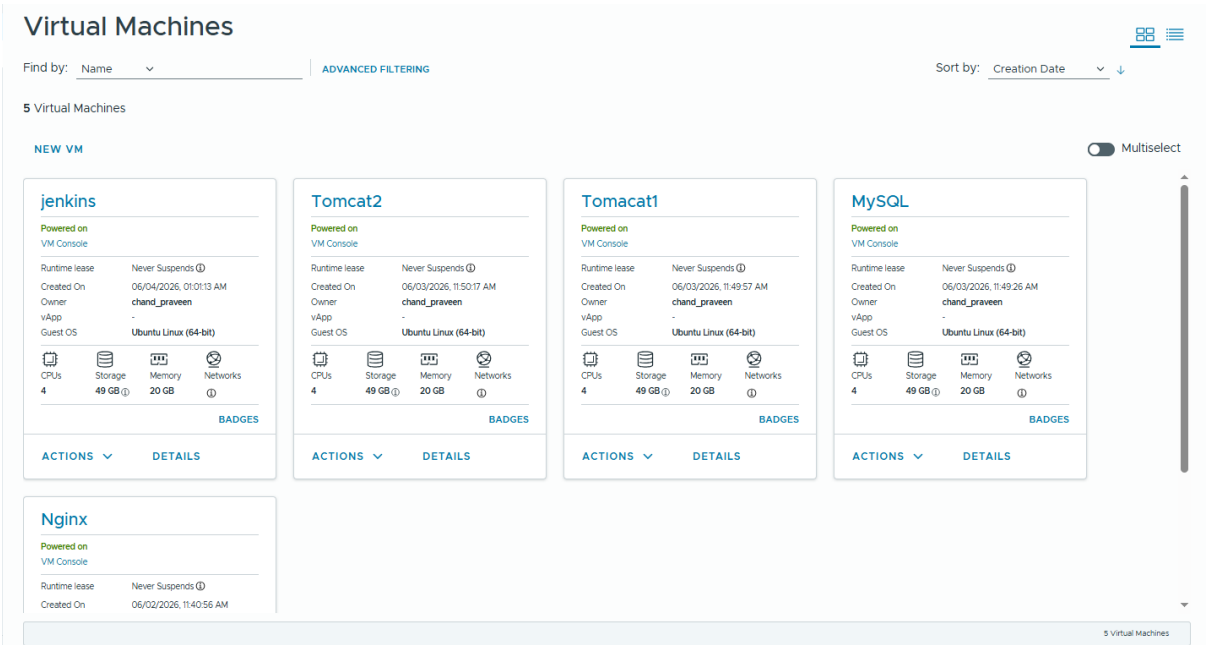


Infrastructure Setup

Description:

Five virtual machines were created to simulate a production environment. Separate VMs were used for Jenkins, MySQL, Nginx Load Balancer, and two Tomcat application servers.



Initial Application Deployment

Before Change

The application was successfully deployed on Tomcat servers and accessed through the Nginx Load Balancer.



NAT Rules

Description:

NAT rules were configured to allow external access to internal virtual machines through public IP addresses.

NAT

RulesLogs

NEWEDITDELETEVIEW LOGS

	Name	Category	State	NAT Action	External IP	Internal IP	Application	External Port	Destination IP	Logging
☐	snatt	User defined	Active	SNAT	49.50.92.88	192.168.0.4	-	Any	-	No
☒	jenkin	User defined	Active	DNAT	49.50.92.81	192.168.0.6	-	Any	-	No
☐	snat	User defined	Active	SNAT	49.50.92.81	192.168.0.6	-	Any	-	No
☐	App1	User defined	Active	DNAT	49.50.92.88	192.168.0.4	-	Any	-	No

Firewall Rules

Description

Firewall rules were implemented to control network traffic between the virtual machines. Only required services such as HTTP, SSH, Jenkins, and database communication were allowed, improving overall security and isolation of the infrastructure.

Firewall

☒ Active

RulesLogs

NEWEDIT RULESEDDITDELETEMOVE TOREARRANGE

	#	Name	Category	State	Applications	Context	Source	Destination
☐	1	jenkins	User defined	Active	general	-	Any	192.168.0.6
☐	2	app2	User defined	Active	general	-	Any	192.168.0.5
☐	3	app1	User defined	Active	general	-	Any	192.168.0.4
☐	4	ssh	User defined	Active	HTTP, HTTPS, 2232 ⓘ	-	Any	192.168.0.2, 192.168.0.4, ...
☐	5	internet	User defined	Active	DNS-TCP, DNS-UDP, H... ⓘ	-	192.168.0.2, 192.168.0.4, ... ⓘ	Any
☐	🔒	default_r...	Default	Active	-	-	Any	Any

Jenkins CI/CD Pipeline

Description:

Jenkins was configured to automatically build and deploy the Java application whenever code changes were pushed to GitHub.

The screenshot shows the Jenkins web interface for a job named 'cicdproject'. The browser address bar indicates the URL is '49.50.92.81:8080/job/cicdproject/'. The Jenkins logo and job name are at the top. A sidebar on the left contains navigation links: Status (selected), Changes, Build Now, Configure, Delete Pipeline, Stages, Rename, Pipeline Syntax, and GitHub Hook Log. The main area shows the job status as 'cicdproject' with a green checkmark icon. Below this is a 'Permalinks' section with a list of build links: Last build (#11), Last stable build (#11), Last successful build (#11), Last failed build (#8), Last unsuccessful build (#8), and Last completed build (#11). At the bottom, a 'Builds' section shows a list of recent builds with a search filter and a table of build details.

Build Number	Time	Status
#12	6:08 AM	Success
#11	7:53 PM	Success
#10	7:40 PM	Success
#9	7:21 PM	Success

Code Modification

Description:

A change was made in the application source code to validate the automated deployment workflow.

```
root@app1: ~/java-login-app
GNU nano 6.2 src/main/webapp/index.jsp *
<html>
<head>
<title>Employee Registration</title>
</head>

<body>

<h1>Hi everyone, This is Login Page</h1>

<h2>Employee Registration</h2>

<form action="register" method="post">

First Name:
<input type="text" name="first_name"><br><br>

Last Name:
<input type="text" name="last_name"><br><br>

Email:

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Git Push

Description:

Modified code was committed and pushed to the GitHub repository.

```
root@app1: ~/java-login-app
Last login: Thu Jun  4 19:38:21 2026 from 223.237.15.34
root@app1:~# cd /root/java-login-app
root@app1:~/java-login-app# nano src/main/webapp/index.jsp
root@app1:~/java-login-app# git add .
git commit -m "Webhook test"
git push origin main
[main 183d983] Webhook test
 1 file changed, 1 insertion(+), 1 deletion(-)
Username for 'https://github.com': Chand643
Password for 'https://Chand643@github.com':
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 526 bytes | 526.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Chand643/loginapp.git
 2bd06d2..183d983  main -> main
root@app1:~/java-login-app# nano src/main/webapp/index.jsp
root@app1:~/java-login-app# git add .
git commit -m "Webhook test"
git push origin main
```

Automatic Jenkins Build

Description:

GitHub Webhook triggered Jenkins automatically. Jenkins fetched the latest code, built the WAR file using Maven, and deployed it to both Tomcat servers.

The screenshot shows the Jenkins web interface for a job named 'cicdproject'. The browser address bar indicates the URL is '49.50.92.81:8080/job/cicdproject/'. The Jenkins logo and job name are at the top. A left sidebar contains navigation links: Status (selected), Changes, Build Now, Configure, Delete Pipeline, Stages, Rename, Pipeline Syntax, and GitHub Hook Log. The main area shows a green checkmark icon and the job name 'cicdproject'. Below this is a 'Permalinks' section with a list of build links: Last build (#11), Last stable build (#11), Last successful build (#11), Last failed build (#8), Last unsuccessful build (#8), and Last completed build (#11), all dated '3 days 10 hr ago'. At the bottom, a 'Builds' section shows a search filter and a list of recent builds. Two builds are visible: build #13 at 6:23 AM and build #12 at 6:08 AM, both marked with green checkmarks. Build #13 has a progress bar and a red 'X' icon.

← ↻ ⚠ Not secure 49.50.92.81:8080/job/cicdproject/

 **Jenkins** / cicdproject

 Status

 Changes

 Build Now

 Configure

 Delete Pipeline

 Stages

 Rename

 Pipeline Syntax

 GitHub Hook Log

 **cicdproject**

Permalinks

- [Last build \(#11\), 3 days 10 hr ago](#)
- [Last stable build \(#11\), 3 days 10 hr ago](#)
- [Last successful build \(#11\), 3 days 10 hr ago](#)
- [Last failed build \(#8\), 3 days 10 hr ago](#)
- [Last unsuccessful build \(#8\), 3 days 10 hr ago](#)
- [Last completed build \(#11\), 3 days 10 hr ago](#)

Builds > ...

🔍 Filter /

Today

	#13 6:23 AM			▼
	#12 6:08 AM			▼

Final Result

Description:

The updated application was successfully deployed and the changes became visible on the live application without manual deployment.



A screenshot of a web browser window. The address bar shows a back arrow, a refresh icon, a warning triangle with the text "Not secure", and the URL "49.50.92.88:8080/loginapp/". The main content area has a large heading "Hi Everyone, This is Login Page" in a bold, black, serif font. Below this is a sub-heading "Employee Registration" in the same font. There are five input fields, each preceded by a label: "First Name:", "Last Name:", "Email:", "Username:", and "Password:". At the bottom left of the form is a button labeled "Register".

← ↻ ⚠ Not secure 49.50.92.88:8080/loginapp/

Hi Everyone, This is Login Page

Employee Registration

First Name:

Last Name:

Email:

Username:

Password: